

Threat Model For Aaronic (oil and Energy) Website Using STRIDE analysis

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INTRODUCTION

This report presents a threat model for Aaronic (Oil and energy) website, outlining potential security risks, vulnerabilities, and mitigation strategies to enhance its security posture.

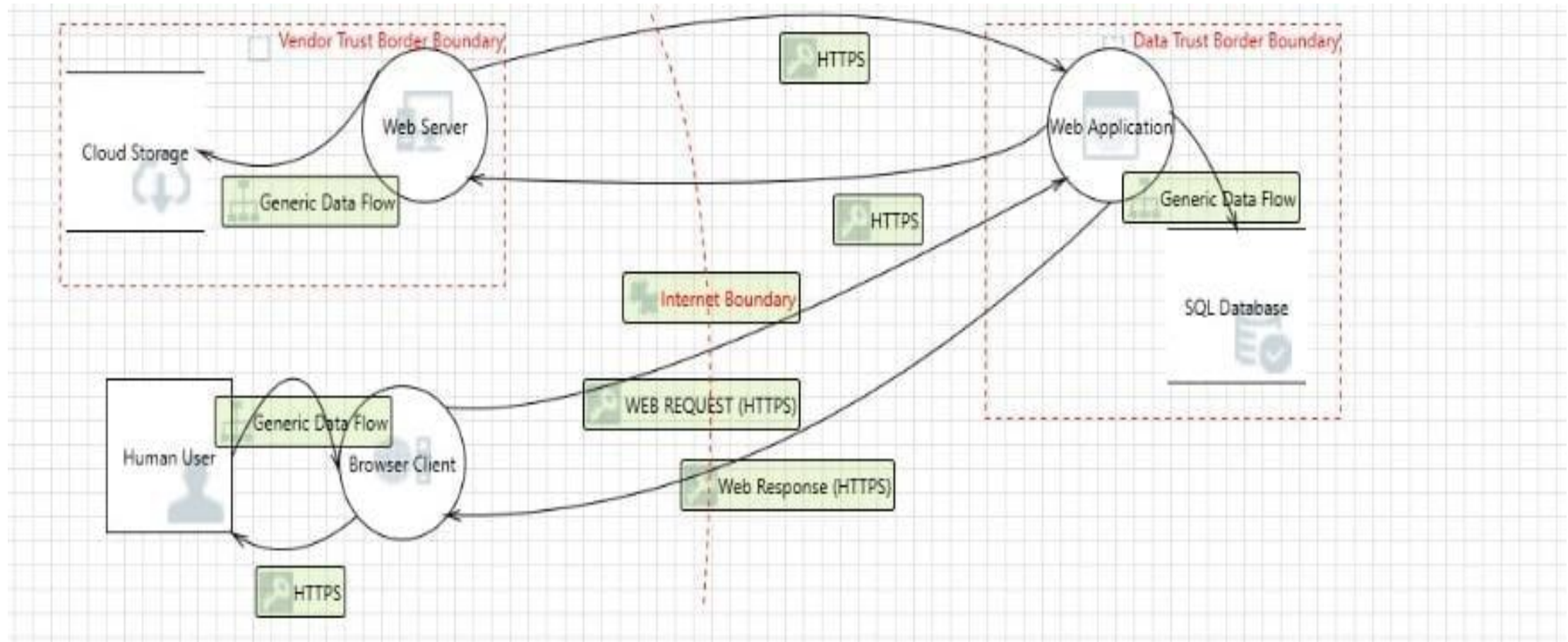
OBJECTIVES

- Identify Potential threats
- Understand the Application Architecture
- Identify and Classify Assets
- Access Vulnerabilities
- Identify threats and Risk levels
- Define Mitigation strategies
- Improve overall security postures

Identified Assets

- **User Data:** vendor information, login credentials for databases.
- **Product Data:** Product details, prices, inventory.
- **Website Availability:** Ensuring the website remains operational and accessible.

Data Flow Diagram



Identified Threats

Component	Threat	Type (STRIDE)	Description
Admin Login	Credential theft	Spoofing	An attacker might steal user credentials through phishing.
Product Pages	Data tampering	Tampering	Attacker can alter databases, files, or records to manipulate information (e.g., changing financial records or grades).
Database	Unauthorized access	Elevation of Privilege	A user might exploit a vulnerability to gain admin rights.
Website	Denial of Service (DoS)	Denial of Service	An attacker might overload the site to make it unavailable.

Identified Vulnerabilities

- Weak password policies allow brute-force attacks.
- Lack of HTTPS encryption exposes data in transit.
- Poor input validation enables SQL Injection attacks.
- Misconfigured access controls may allow unauthorized database access.

Risk Evaluation

This Risk Evaluation Matrix assesses cybersecurity threats based on their likelihood, impact, and overall risk level.

Credential theft is the most critical threat due to its high likelihood and high impact.

Unauthorized access has a low likelihood but a critical impact, making it a high-risk concern.

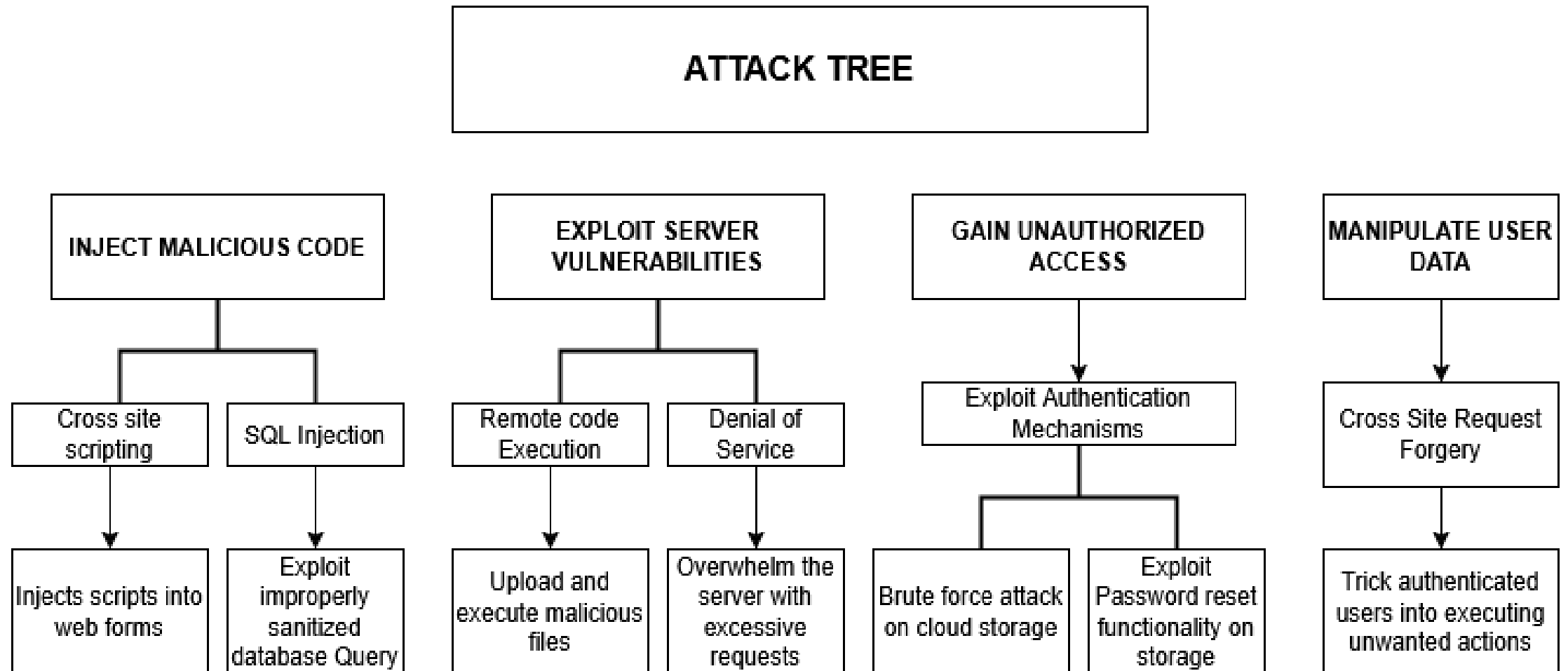
Denial of Service (DoS) and Data tampering are both medium-likelihood, high-impact threats, ranking as high risks.

Data leakage is a lower-priority risk with low likelihood and medium impact.

Overall, credential theft and unauthorized access pose the most significant risks, requiring strong mitigation strategies.

Threat	Likelihood	Impact	Risk Level
Credential theft	High	High	Critical
Data tampering	Medium	High	High
Unauthorized access	Low	Critical	High
Denial of Service (DoS)	Medium	High	High
Data leakage	Low	Medium	Medium

Attack Tree



Mitigation Strategies

Threat	Mitigation Strategy
Credential theft	Implement multi-factor authentication (MFA) and strong password policies.
Data tampering	Use HTTPS to encrypt data in transit.
Unauthorized access	Implement role-based access control (RBAC) and security audits.
Denial of Service (DoS)	Use rate limiting and DDoS protection services.
Data leakage	Encrypt sensitive data at rest and restrict data access.

Conclusion

- This threat model provides an in-depth analysis of the security threats facing Aaronic (Oil and Energy) website and proposes actionable mitigation strategies. Regular security audits, penetration testing, and continuous monitoring should be implemented to ensure the website remains protected against emerging threats.